Spotify Data Analysis

Unlocking the Recipe for a Hit

An analysis of audio features, genre trends, and track popularity.

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Brief History of Spotify

Origins & Scale

- **Foundation:** Founded in **2006** by **Daniel Ek** and **Martin Lorentzon** in **Stockholm, Sweden**.
- **Launch:** The service officially launched in **October 2008.**
- **Scale:** Offers music streaming with over 70 million tracks.

Key Services

- **Discovery:** Curated playlists like **"Discover Weekly"** and **"Release Radar"**.
- **Podcasts:** Hosts **exclusive content** and popular shows.
- **Personalization:** Algorithms provide recommendations based on user habits.
- **Accessibility:** Available cross-platform on smartphones, smart speakers and desktops.

Source and Description of Dataset

Source

The dataset was obtained from **Kaggle** and contains data originally extracted via the **Spotify Web API**.

Description

The dataset comprises numerous rows, each representing a unique track.

It includes attributes such as **track name, artist, popularity score, genre** and **detailed audio features**.

Glossary (Key Audio Features)

Valence

Describes musical "**positiveness**." High values sound happy; low values sound sad.

Energy

Represents **intensity** and **activity**. High energy tracks feel fast and loud.

Danceability

Describes how suitable a track is for **dancing** based on **tempo** and **beat** strength.

Acousticness

Confidence measure of whether the track is **acoustic** (unplugged).

Liveness

Detects the **presence of an audience** (e.g., live concert recordings).

Loudness

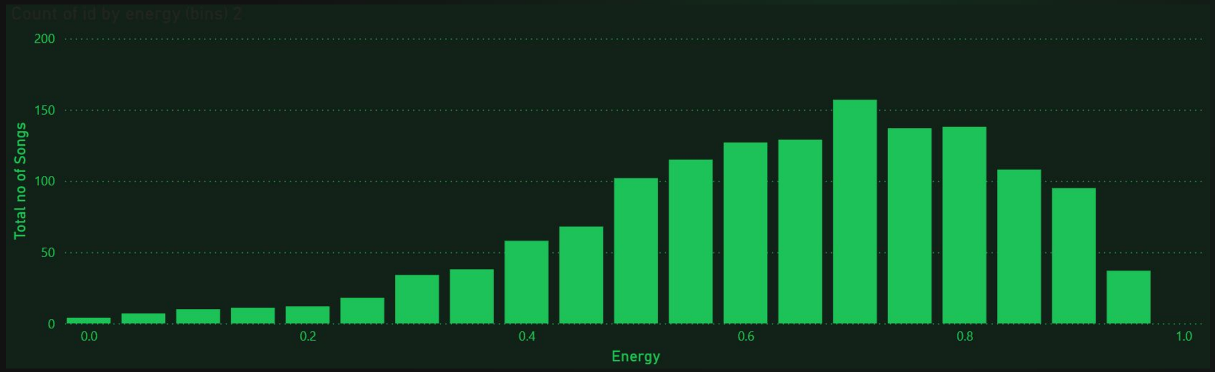
The **overall volume of a track** in decibels (dB).

Music Intensity Analysis

Energy Levels

The data is skewed heavily toward high energy. Most tracks fall between **0.6 and 0.9** on the energy scale.

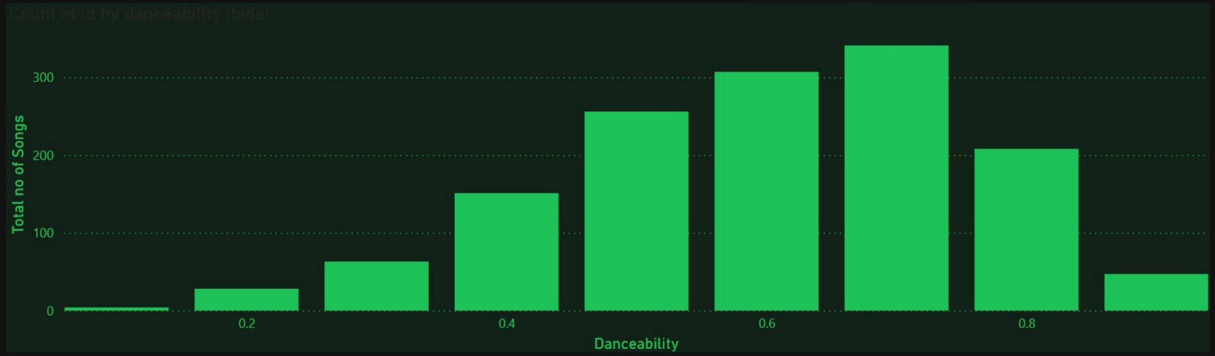
This indicates a platform preference for dynamic, vibrant tracks often found in Rock and Pop.



Dance Potential

Similar to energy, **most tracks score high** on danceability.

Tracks with high danceability are ideal for social environments and parties.



Mood & Speed Breakdown

Emotional Vibe

Valence is distributed more evenly than energy, though there is a slight preference for **neutral-to-positive** moods. This variety allows Spotify to cater to diverse emotional states , from focus (low valence) to party (high valence).

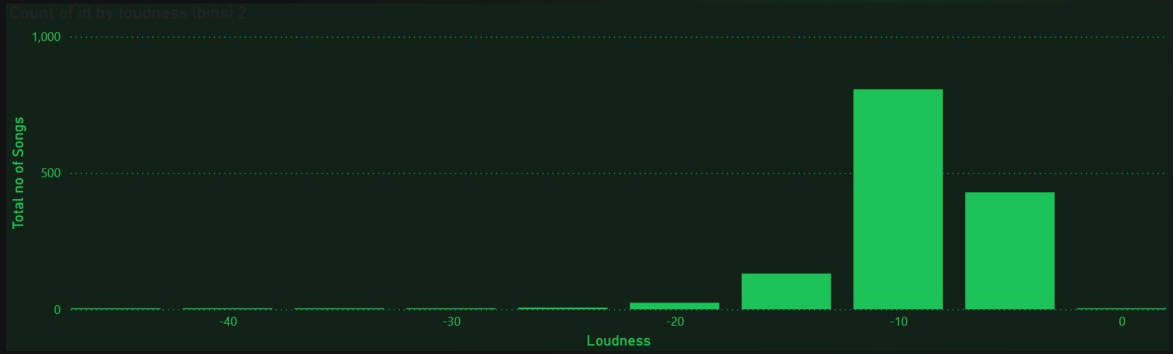
Track Speed

There is a massive spike in tracks **around 120-130 BPM**. This is the standard tempo for **Pop** and **Dance music**, making these genres dominant on the platform.

Sound Volume & Recording Type

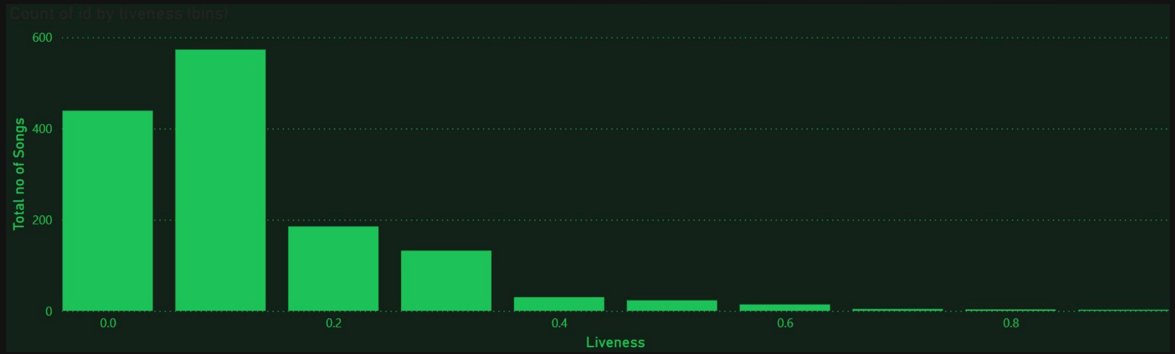
Track Volume

The distribution is **"Left-Skewed"** meaning **the vast majority of tracks are very loud (between -10dB and -5dB)**. This reflects modern mastering standards **("The Loudness Wars") in Pop, Metal, and Electronic music**.

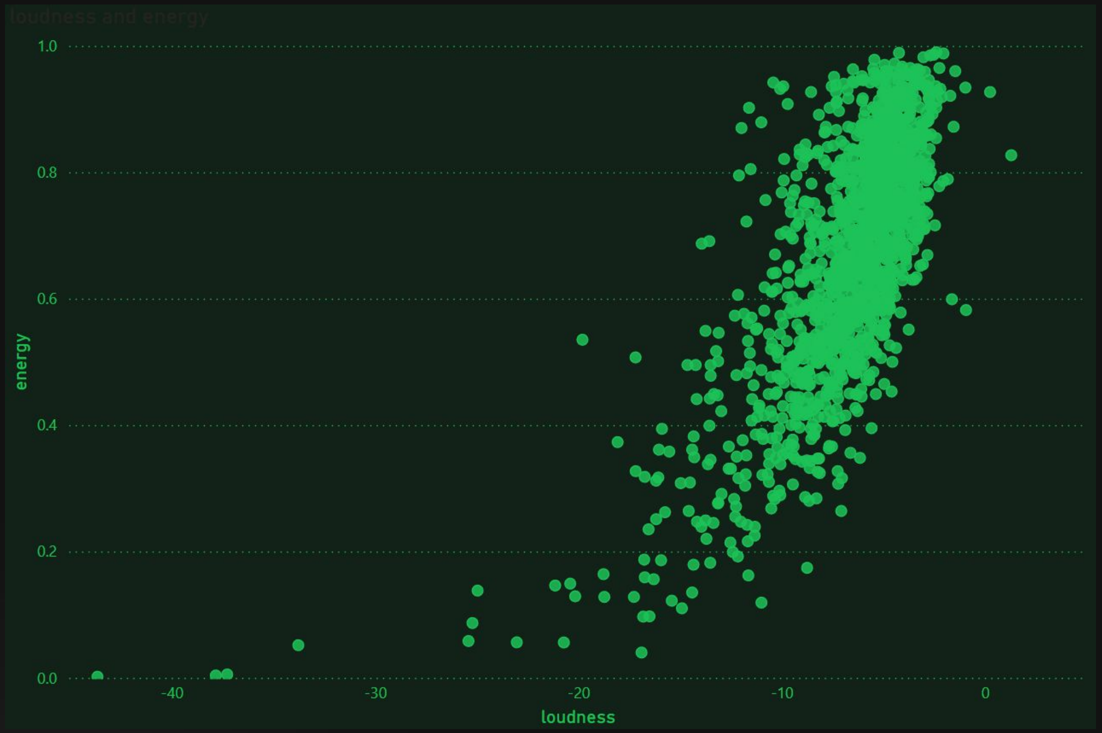


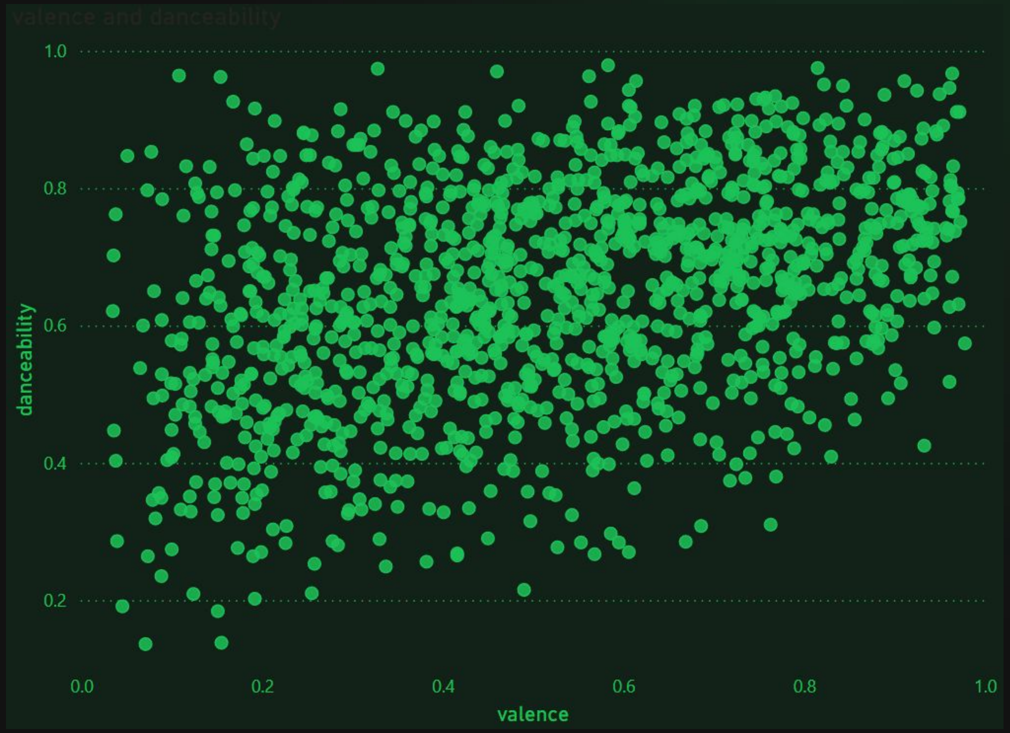
Live vs. Studio Recording

The distribution is heavily **"Right-Skewed"** Most tracks have **near-zero liveness**. This confirms that Spotify is primarily **a library of studio-recorded music**, with very **few live concert recordings**.



Audio Features Correlation Analysis





Loudness & Energy (0.71): As expected, louder tracks are almost always more energetic.

Valence & Danceability (0.37): Happy songs tend to be more danceable.

Most Popular Genres

"**Pop** is the undisputed leader, significantly outpacing all other categories. This suggests that the average listener gravitates heavily toward mainstream hits, though the strong showing for **Rock** and **Hip-Hop** proves **there is still significant demand for classic and rhythmic styles.**"

Correlation with Popularity

The Myth

It is often assumed that louder or faster songs are automatically more popular.

We expect technical perfection to drive hits.

The Reality

Data shows **no significant correlation** between audio features and popularity.

- **Loudness: 0.03 (Negligible)**
- **Liveness: -0.026 (Negligible)**
- **Energy: 0.001 (Negligible)**

Conclusion: *A track's technical composition does not determine its success.*

Playlist Profiling

Playlist Identities: Distinct playlists have distinct audio "fingerprints".

Global Top 50

Features **high Valence (Average ~0.51)**, indicating a preference for **cheerful, upbeat** hits globally.

Today's Top Hits

Surprisingly features higher **Acousticness (>0.20)** compared to Gaming, suggesting a trend toward **"Pop-Acoustic"** vocals.

Top Gaming Tracks

Marked by **high energy** but **lower acousticness, fitting the focus required for gaming.**

Key Results & Insights

- **The "Spotify Sound"** The platform is dominated by **high-energy**, **high-danceability** and **high-loudness tracks**.
- **Audio Doesn't Drive Hits** There is no "magic formula" for audio features. **A sad, slow song has just as much chance of being a hit as a fast, loud one**.
- **Recency is King** The strongest driver of popularity is the release year. Users predominantly engage with content released in the **last 5-10 years**.
- **Genre Matters** **Pop and Latin** genres have a higher baseline probability of reaching the top charts than niche genres.

Key Takeaway

Algorithms

Since users favor recency, algorithms should prioritize new releases while using audio features only to match specific user moods.

Artist Strategy

Focus on frequent releases to leverage **"Recency Bias"** and produce songs to fit specific audio profiles of target playlists.

Marketing

Promotional efforts should focus on the Pop and Latin markets, as these yield the highest volume of **"Super-Hits"**.